

Problems on Numbers

1. The sum of two numbers is 16 and the sum of their squares is 130. The numbers are:

A) 6, 10

B) 7, 9

C) 5, 11

D) 4, 12

Answer: B) 7, 9

2. The product of two consecutive odd numbers is 195. The smaller number is:

A) 11

B) 13

C) 15

D) 17

Answer: B) 13

3. A number is multiplied by 4 and then 12 is added to it. The result is 60. The number is:

A) 10

B) 12

C) 14

D) 16

Answer: B) 12

4. If the difference between a number and its reciprocal is $\frac{5}{6}$, the number is:

A) $\frac{2}{3}$

B) $\frac{3}{2}$

C) $\frac{4}{3}$

D) $\frac{5}{2}$

Answer: B) $\frac{3}{2}$

5. The sum of three consecutive natural numbers is 87. The middle number is:

A) 27

B) 28

C) 29

D) 30

Answer: C) 29

6. Divide 80 into two parts such that the greater part exceeds twice the smaller by 11.

A) 55, 25

B) 57, 23

C) 52, 28

D) 50, 30

Answer: B) 57, 23

7. The difference between the squares of two numbers is 25600 and the sum of the numbers is 1000. The numbers are:

A) 600, 400

B) 640, 360

C) 512.8, 487.2

D) 512, 488

Answer: C) 512.8, 487.2 (Wait, let's recheck integer options: $x^2 - y^2 = 25600$, $x + y = 1000 \rightarrow x - y = 25.6$. $x = 512.8$. Let's provide integer friendly question).

Correction for Q7: The difference between the squares of two numbers is 45 and their sum is 9. The numbers are:

A) 5, 4

B) 6, 3

C) 7, 2

D) 8, 1

Answer: C) 7, 2

8. A number exceeds its one-fifth by 20. The number is:

A) 20

B) 25

C) 30

D) 35

Answer: B) 25

9. If one-seventh of a number exceeds its one-eleventh by 100, then the number is:

A) 770

B) 1100

C) 1825

D) 1925

Answer: D) 1925

10. The sum of a number and its square is 240. The number is:

A) 12

B) 14

C) 15

D) 16

Answer: C) 15

11. Find a number such that 4 times the number exceeds half the number by 49.

A) 12

B) 14

C) 16

D) 18

Answer: B) 14

12. The sum of two numbers is 20 and their product is 96. The difference of the numbers is:

A) 4

B) 8

C) 2

D) 6

Answer: A) 4

13. Three numbers are in the ratio 3:4:5 and the sum of their squares is 1250. The sum of the numbers is:

- A) 30
- B) 50
- C) 60
- D) 90

Answer: C) 60

14. The product of two numbers is 9375 and the quotient, when the larger one is divided by the smaller, is 15. The sum of the numbers is:

- A) 380
- B) 395
- C) 400
- D) 425

Answer: C) 400

15. The sum of the digits of a two-digit number is 11. If 27 is subtracted from the number, the digits reverse their order. The number is:

- A) 83
- B) 74
- C) 65
- D) 92

Answer: B) 74

16. Which number is 60% less than 80?

- A) 48
- B) 32
- C) 20
- D) 24

Answer: B) 32

17. A number when added to its $\frac{2}{3}$ rd makes 45. The number is:

A) 27

B) 30

C) 36

D) 18

Answer: A) 27

18. If a number is decreased by 4 and divided by 6, the result is 8. The number is:

A) 48

B) 50

C) 52

D) 54

Answer: C) 52

19. The sum of two numbers is 29 and the difference of their squares is 145. The difference between the numbers is:

A) 3

B) 5

C) 8

D) 11

Answer: B) 5

20. $\frac{2}{3}$ of a number is less than $\frac{3}{4}$ of the number by 6. The number is:

A) 36

B) 48

C) 72

D) 84

Answer: C) 72

21. The product of two numbers is 108 and the sum of their reciprocals is $\frac{3}{12}$. (Correction: Let's assume sum of reciprocals is $\frac{7}{36}$).

Revised Q21: The sum of two numbers is 12 and their product is 35. The sum of their reciprocals is:

A) $12/35$

B) $35/12$

C) $7/12$

D) $5/7$

Answer: A) $12/35$

22. A number consists of two digits. The digit in the ten's place exceeds the digit in the unit's place by 4. The sum of the digits is $1/7$ of the number. The number is:

A) 84

B) 62

C) 73

D) 48

Answer: A) 84

23. Which number must be added to each of the numbers 6, 15, 20 and 43 to make them proportional?

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

24. A number is such that $1/3$ of it is equal to $1/4$ of another number. If the sum of the two numbers is 77, find the smaller number.

A) 22

B) 33

C) 44

D) 55

Answer: B) 33

25. The sum of five consecutive even numbers is 200. The smallest of these numbers is:

A) 34

B) 36

C) 38

D) 40

Answer: B) 36

26. 45 is four times a number minus 3. The number is:

A) 10

B) 11

C) 12

D) 13

Answer: C) 12

27. The difference between a number and its $\frac{2}{7}$ th is 30. The number is:

A) 35

B) 42

C) 49

D) 56

Answer: B) 42

28. The sum of two numbers is 28. If one of them is $\frac{3}{4}$ of the other, the larger number is:

A) 12

B) 14

C) 16

D) 18

Answer: C) 16

29. Find a number such that the ratio of the number to 12 is the same as the ratio of 3 to the number.

A) 4

B) 6

C) 8

D) 9

Answer: B) 6

30. If the sum of a number and its square is 156, the number is:

A) 11

B) 12

C) 13

D) 14

Answer: B) 12

31. If you subtract -2 from a number and multiply the result by -5, you get 20. The number is:

A) -6

B) 2

C) 6

D) -2

Answer: A) -6

32. The sum of the squares of three consecutive natural numbers is 110. The numbers are:

A) 5, 6, 7

B) 6, 7, 8

C) 4, 5, 6

D) 7, 8, 9

Answer: A) 5, 6, 7

33. A two-digit number is 3 times the sum of its digits. If 45 is added to it, the digits are reversed. The number is:

A) 27

B) 72

C) 23

D) 32

Answer: A) 27

34. A number when divided by 119 leaves a remainder of 19. If it is divided by 17, the remainder will be:

A) 2

B) 3

C) 7

D) 10

Answer: A) 2

35. The sum of two numbers is 100 and their difference is 37. The difference of their squares is:

A) 37

B) 100

C) 6300

D) 3700

Answer: D) 3700

36. $\frac{1}{4}$ of a number subtracted from $\frac{1}{3}$ of the number gives 12. The number is:

A) 144

B) 120

C) 72

D) 60

Answer: A) 144

37. If x is an even number, which of the following is always odd?

A) $x + 2$

B) $2x$

C) x^2

D) $x + 1$

Answer: D) $x + 1$

38. The sum of three consecutive multiples of 5 is 60. The largest number is:

A) 15

B) 20

C) 25

D) 30

Answer: C) 25

39. A number is multiplied by $\frac{3}{4}$ and the result is 60. The number is:

A) 45

B) 60

C) 80

D) 90

Answer: C) 80

40. Two numbers are in the ratio 4:7. If their sum is 1331, find the smaller number.

A) 484

B) 500

C) 600

D) 847

Answer: A) 484

41. The difference between two positive integers is 3. The sum of their squares is 369. The numbers are:

A) 12, 15

B) 11, 14

C) 13, 16

D) 8, 11

Answer: A) 12, 15

42. If a number is increased by 25%, it becomes 125. The number is:

- A) 80
- B) 90
- C) 100
- D) 110

Answer: C) 100

43. The product of two numbers is 72. If one number is twice the other, find the larger number.

- A) 6
- B) 12
- C) 8
- D) 18

Answer: B) 12

44. A number is less than 50 and is a multiple of 7. The sum of its digits is 10. The number is:

- A) 28
- B) 35
- C) 42
- D) 49

Answer: A) 28

45. The sum of 4 consecutive even numbers is 156. The largest number is:

- A) 38
- B) 40
- C) 42
- D) 44

Answer: C) 42

46. What number should be subtracted from both the numerator and denominator of $\frac{15}{19}$ so that it becomes $\frac{3}{4}$?

- A) 3
- B) 4

C) 5

D) 6

Answer: A) 3

47. The quotient of two numbers is 4 and their sum is 35. The larger number is:

A) 20

B) 24

C) 28

D) 30

Answer: C) 28

48. A number divided by 68 gives the quotient 269 and remainder 0. The number is:

A) 18292

B) 18362

C) 18452

D) 18522

Answer: A) 18292

49. Three numbers are in the ratio 1:2:3. The sum of their cubes is 4500. The smallest number is:

A) 4

B) 5

C) 6

D) 10

Answer: B) 5

50. $\frac{4}{7}$ of a number is 24 less than the number itself. The number is:

A) 42

B) 49

C) 56

D) 63

Answer: C) 56

51. The sum of a number and its square is 110. The number is:

- A) 10
- B) 11
- C) 9
- D) 8

Answer: A) 10

52. The product of two consecutive even numbers is 168. The sum of the numbers is:

- A) 24
- B) 26
- C) 28
- D) 30

Answer: B) 26

53. If the square of a number is added to the square of 12, the result is 225. The number is:

- A) 8
- B) 9
- C) 10
- D) 11

Answer: B) 9

54. The value of a number is 6 times the value of another. Their sum is 49. The larger number is:

- A) 36
- B) 42
- C) 45
- D) 48

Answer: B) 42

55. Find a number which is less than 100 and is divisible by 9, and the difference of its digits is 7.

- A) 90

B) 92

C) 18

D) 29

Answer: A) 90 (Sum=9, Diff=9), Wait. Let's re-eval. Divisible by 9: 09, 18, 27, 36, 45, 54, 63, 72, 81, 90.

Diff 7: $7-0=7$ (no), $8-1=7$ (18 is yes!), $9-2=7$ (29 no). $1-8=7$.

Let's check options. 18: diff is 7. 90: diff is 9.

Answer: C) 18

56. The sum of two numbers is 15. The sum of their reciprocals is $\frac{3}{10}$. The numbers are:

A) 5, 10

B) 6, 9

C) 7, 8

D) 3, 12

Answer: A) 5, 10

57. A number is 50 more than 50% of itself. The number is:

A) 50

B) 75

C) 100

D) 150

Answer: C) 100

58. $\frac{3}{8}$ of a number is 9. What is $\frac{1}{2}$ of that number?

A) 12

B) 16

C) 18

D) 24

Answer: A) 12

59. The difference between two numbers is 2. The difference of their cubes is 56. The numbers are:

A) 3, 5

B) 4, 2

C) 6, 4

D) 5, 3

Answer: B) 4, 2

60. The sum of 6 consecutive odd numbers is 120. The largest number is:

A) 23

B) 25

C) 27

D) 29

Answer: B) 25

61. Two numbers are in ratio 2:3. If 2 is added to the first and 2 is subtracted from the second, the ratio becomes 5:6. (Check: $(2x+2)/(3x-2)=5/6 \rightarrow 12x+12=15x-10 \rightarrow 3x=22 \rightarrow x=7.33$. Not integer).

Revised Q61: Two numbers are in the ratio 5:8. If 12 is added to each, they are in the ratio 3:4. The smaller number is:

A) 15

B) 20

C) 25

D) 30

Answer: A) 15

62. If a number is multiplied by $\frac{2}{3}$ and then 15 is subtracted, the result is 25. The number is:

A) 45

B) 60

C) 75

D) 90

Answer: B) 60

63. The sum of the squares of two natural numbers is 41 and the sum of the numbers is 9. The product of the numbers is:

A) 15

B) 18

C) 20

D) 24

Answer: C) 20

64. What is the number which when multiplied by 13 increases by 144?

A) 10

B) 11

C) 12

D) 13

Answer: C) 12

65. A number consists of two digits. The digit at the unit's place is double the digit at the ten's place. The number exceeds the sum of its digits by 18. The number is:

A) 24

B) 36

C) 48

D) 12

Answer: A) 24

66. $\frac{1}{5}$ of a number exceeds $\frac{1}{7}$ of the same number by 10. The number is:

A) 125

B) 150

C) 175

D) 200

Answer: C) 175

67. The product of two numbers is 5043 and their H.C.F is 41. The larger number is:

A) 41

B) 123

C) 164

D) 205

Answer: B) 123

68. Find the positive number which when decreased by 20 is equal to 69 times the reciprocal of the number.

A) 21

B) 23

C) 25

D) 27

Answer: B) 23

69. The sum of two numbers is 45 and their ratio is 7:8. The numbers are:

A) 20, 25

B) 21, 24

C) 22, 23

D) 18, 27

Answer: B) 21, 24

70. If 40 is added to 60% of a number, the result is the number itself. The number is:

A) 80

B) 90

C) 100

D) 120

Answer: C) 100

71. The sum of a number and its $\frac{1}{3}$ rd is 64. The number is:

A) 42

B) 48

C) 52

D) 56

Answer: B) 48

72. If the difference of two numbers is 3 and the difference of their squares is 39, find the larger number.

- A) 8
- B) 9
- C) 12
- D) 13

Answer: A) 8

73. A number divided by 5 leaves a remainder 3. What is the remainder when the square of the same number is divided by 5?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: D) 4

74. The sum of 3 consecutive even numbers is 18 more than twice the largest. The smallest number is:

- A) 10
- B) 12
- C) 14
- D) 16

Answer: B) 12

75. $\frac{3}{5}$ of $\frac{1}{4}$ of a number is 18. The number is:

- A) 110
- B) 120
- C) 130
- D) 140

Answer: B) 120

76. The product of two fractions is $\frac{14}{15}$ and their quotient is $\frac{35}{24}$. The greater fraction is:

A) $\frac{7}{6}$

B) $\frac{7}{4}$

C) $\frac{7}{3}$

D) $\frac{4}{5}$

Answer: A) $\frac{7}{6}$

77. The sum of the digits of a 3-digit number is 12. (Incomplete or ambiguous without more conditions. Let's replace.)

Revised Q77: A number when divided by 12 leaves a remainder 7. When the same number is divided by 3, the remainder is:

A) 0

B) 1

C) 2

D) 3

Answer: B) 1

78. If 1 is added to the numerator of a fraction, it becomes $\frac{1}{2}$. If 1 is added to the denominator, it becomes $\frac{1}{3}$. The fraction is:

A) $\frac{4}{11}$

B) $\frac{5}{12}$

C) $\frac{2}{9}$

D) $\frac{3}{10}$

Answer: A) $\frac{4}{11}$

79. The sum of two numbers is 60 and their difference is 10. The difference of their squares is:

A) 500

B) 600

C) 700

D) 800

Answer: B) 600

80. Which of the following numbers is divisible by 3, 4, 5 and 6?

- A) 36
- B) 60
- C) 80
- D) 90

Answer: B) 60

81. The square of a positive number is 2500 more than the square of 10. The number is:

- A) 50
- B) 51
- C) 52
- D) 60

Answer: B) 51 (Wait, $51^2 = 2601$. $10^2 = 100$. $2601 - 100 = 2501$. My calculation was fast. Let's check $50^2 = 2500$. $2500 - 100 = 2400$. Let's solve $x^2 - 100 = 2500 \rightarrow x^2 = 2600$. $\sqrt{2600}$ is not integer. Let's fix question).

Correction Q81: The difference between the square of a number and the square of 20 is 500. The number is:

- A) 25
- B) 30
- C) 35
- D) 40

Answer: B) 30

82. $\frac{5}{6}$ of a number is 25. The number is:

- A) 20
- B) 25
- C) 30
- D) 35

Answer: C) 30

83. The sum of the digits of a two-digit number is 10. If the digits are reversed, the new number increases by 18. The number is:

A) 46

B) 64

C) 28

D) 82

Answer: A) 46

84. What is the number which when divided by 7 gives a quotient 15 and a remainder 4?

A) 105

B) 109

C) 115

D) 119

Answer: B) 109

85. Three numbers are such that the second is twice the first and half the third. If the sum of the three numbers is 56, the smallest number is:

A) 8

B) 16

C) 24

D) 32

Answer: A) 8

86. The product of two numbers is 2500. If the first number is 4 times the second, the numbers are:

A) 25, 100

B) 50, 50

C) 10, 250

D) 20, 125

Answer: A) 25, 100

87. A number exceeds 25 by as much as it is less than 55. The number is:

A) 35

B) 38

C) 40

D) 45

Answer: C) 40

88. If 30% of a number is 12.6, then the number is:

A) 41

B) 42

C) 45

D) 48

Answer: B) 42

89. A number whose double is 45 greater than its half is:

A) 20

B) 25

C) 30

D) 40

Answer: C) 30

90. The sum of two numbers is 50. If the larger number is divided by the smaller, the quotient is 4. The numbers are:

A) 10, 40

B) 15, 35

C) 20, 30

D) 5, 45

Answer: A) 10, 40

91. $\frac{2}{5}$ of a number is 14. What is 150% of that number?

A) 35

B) 52.5

C) 70

D) 105

Answer: B) 52.5

92. The product of two consecutive odd integers is 63. The sum of the integers is:

A) 12

B) 14

C) 16

D) 18

Answer: C) 16

93. A number consists of two digits. The sum of the digits is 6. If 18 is subtracted from the number, the digits are reversed. The number is:

A) 42

B) 24

C) 51

D) 15

Answer: A) 42

94. $\frac{3}{4}$ of a number is 21. What is the number?

A) 24

B) 26

C) 28

D) 32

Answer: C) 28

95. If 5 is added to twice a number, the result is 29. The number is:

A) 10

B) 12

C) 14

D) 16

Answer: B) 12

96. The sum of two numbers is 8 and the sum of their squares is 34. The product of the numbers is:

A) 12

B) 15

C) 16

D) 20

Answer: B) 15

97. If a number is multiplied by 5 and 10 is subtracted, the result is 30. The number is:

A) 6

B) 8

C) 10

D) 12

Answer: B) 8

98. The ratio of two numbers is 5:6 and their sum is 55. The difference is:

A) 5

B) 6

C) 11

D) 1

Answer: A) 5

99. 0.6 of a number is equal to 0.09 of another number. The ratio of the numbers is:

A) 2:3

B) 3:20

C) 20:3

D) 1:15

Answer: B) 3:20

100. The sum of three consecutive even numbers is 54. The largest number is:

A) 16

B) 18

C) 20

D) 22

Answer: C) 20